INTERACTION DESIGN BASICS

Outline

- Design:
  - what it is, interventions, goals, constraints
- The design process
  - what happens when
- Users
  - who they are, what they are like …
- Scenarios
  - rich stories of design
- Navigation
  - finding your way around a system
- Iteration and prototypes
  - never get it right first time!

What is design?

Achieving goals within constraints

- Goals - purpose
  - who is it for, why do they need it
- Constraints
  - materials, platforms
- Trade-offs

What is design?

- "It is where you stand with a foot in two worlds –the world of technology and the world of people to do human purposes – and you try to bring the two for together" (*Mitch Kapor 1996*)

Definition of Design

- The term ‘design’ refers to:
  - It is the creative process of specifying something new.
  - And the representations that are produced during the process.

Good and bad design

- What is wrong with the remote on the right?
- Why is the TiVo remote so much better designed?
  - Peanut shaped to fit in hand
  - Logical layout and color-coded, distinctive buttons
  - Easy to locate buttons

TiVo

Golden rule of design

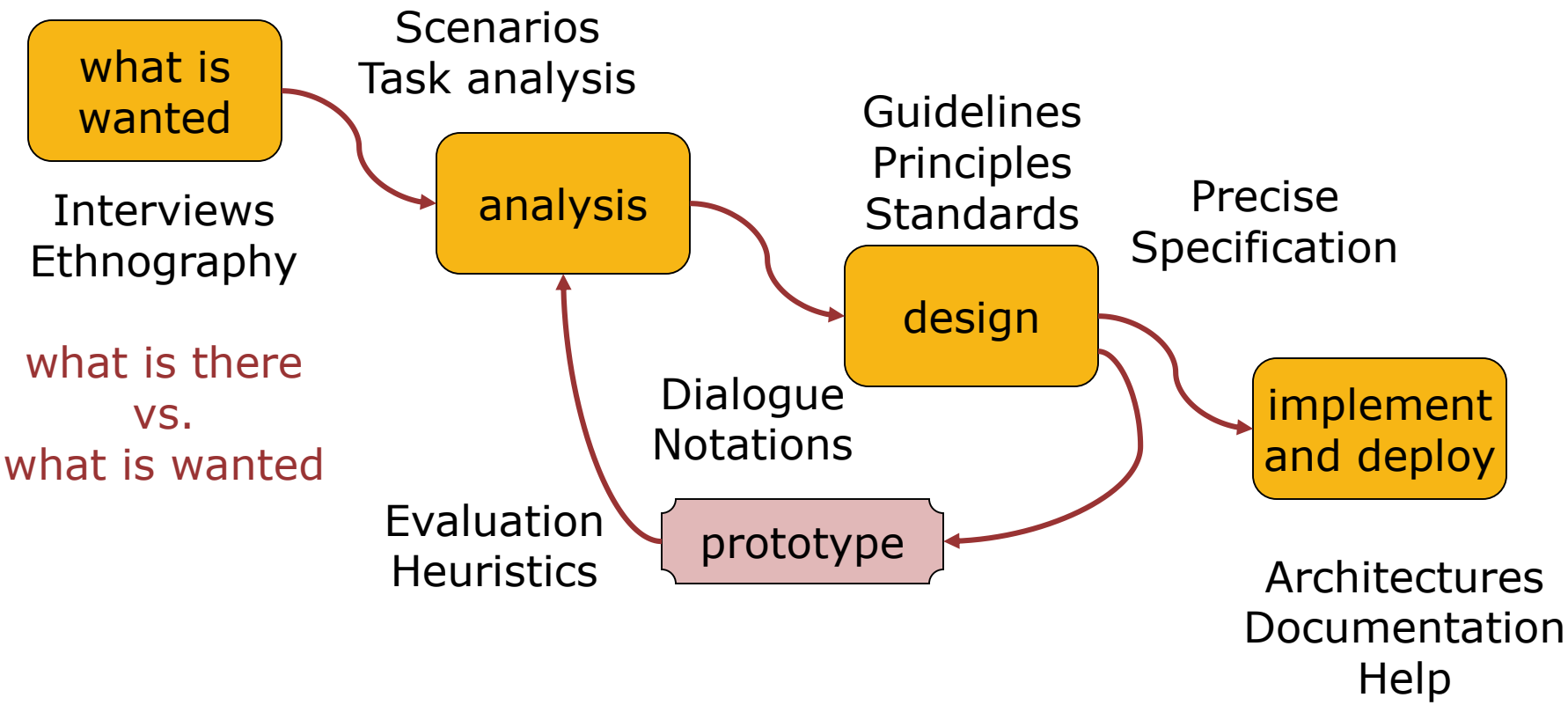
Understand your materials

For Human–Computer Interaction

Understand your materials

- Understand Computers
  - limitations, capacities, tools, platforms
- Understand People
  - psychological, social aspects
  - human error
- And their interaction …

The process of design



Ethnography

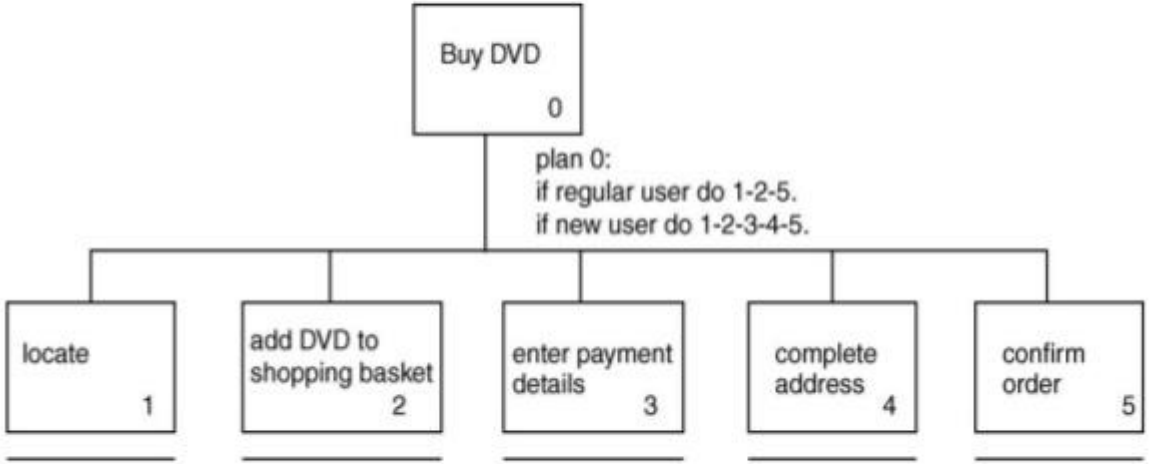
- **Ethnography** (from Greek *ethnos* " folk, people, nation" and *grapho* "I write") is the systematic study of people and cultures.
- It is the scientific description of peoples and cultures with their customs, habits, and mutual differences.

Task Analysis

- Helps us understand what users do
- Forces us to synthesize the data into a useful format
- Lays the foundation for screen design
- Identifies weak points and sources of error in the task flow

Example for Task Analysis

Example Hierarchical Task Analysis (graphical)



Scenario

Scenario - Example

- A scenario is a little story
- It describes one possible course of events:

- Customer puts ATM card into machine and types password
- System validates the card and password
- System prompts the customer for a transaction type and amount
- Customer selects "withdraw $100 from checking"
- System checks with the central bank make sure customer has sufficient funds
- System dispenses the cash, returns card, and prints receipt
- System asks customer if he/she would like another transaction

Steps …

- Requirements
  - what is there and what is wanted …
- Analysis
  - ordering and understanding
- Design
  - what to do and how to decide
- Iteration and prototyping
  - getting it right … and finding what is really needed!
- Implementation and deployment
  - making it and getting it out there

USER FOCUS

know your user

personae

cultural probes

know your user

- Who are they?
- Probably <u>not</u> like you!
- Talk to them
- Watch them
- Use your imagination